

Module 3: Visualize & Analyze the Data

Module 3 – Learning Objectives



1. Get familiar with different types of visualizations in Power BI Desktop



2. Create reports in Power BI Desktop



3. Publish reports created in Power BI Desktop to the Power BI Service

Module 3 – Outline

1. Lesson 1: Exploring the Visualization Gallery & Choosing Appropriate Visuals in Power BI Desktop
2. Lesson 2: Creating Reports
3. Lesson 3: Formatting Reports
4. Lesson 4: Publishing a Report to the Power BI Service

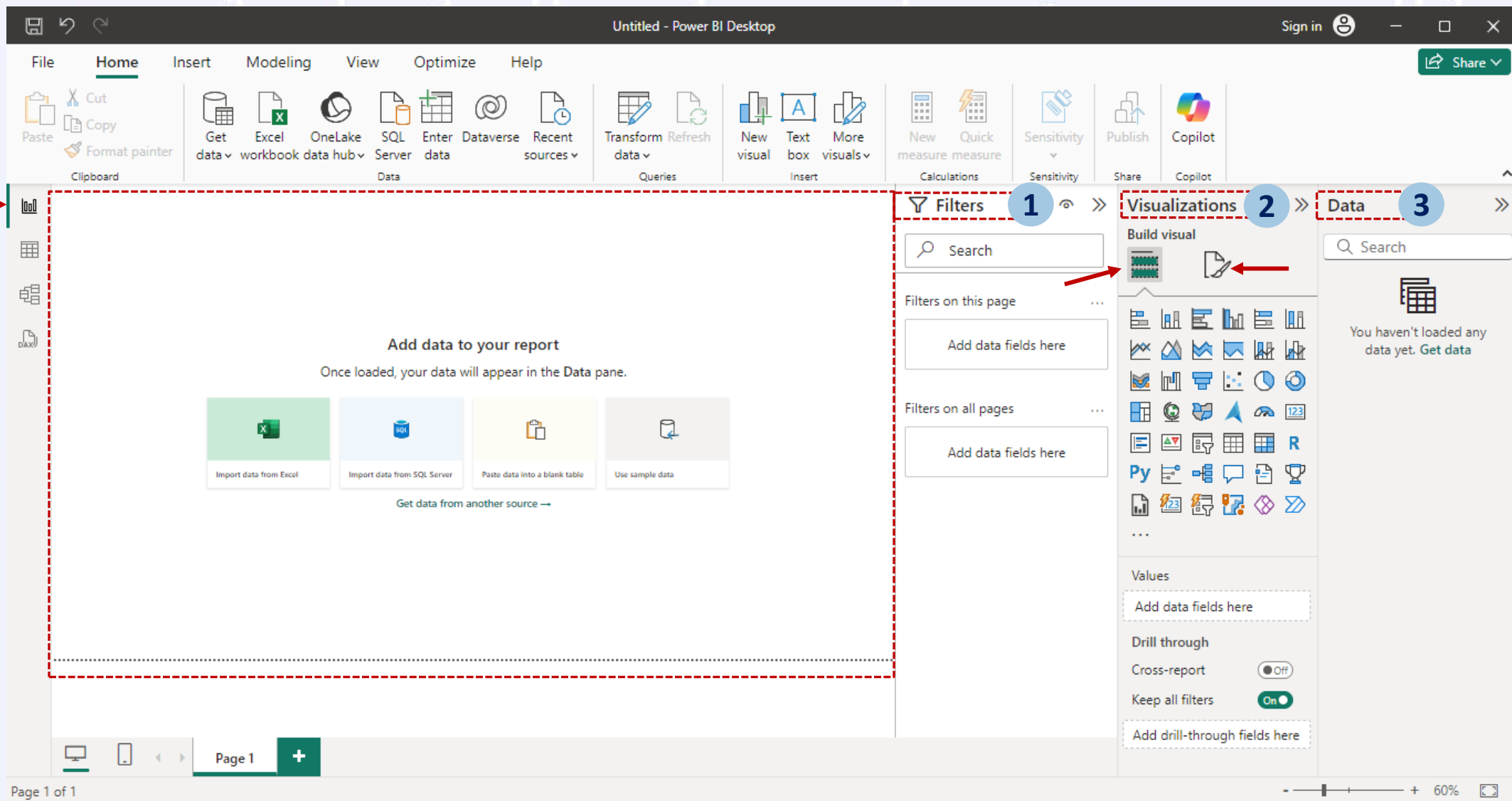
Lesson 1 – Exploring the Visualization Gallery & Choosing Appropriate Visuals in Power BI Desktop

Objectives:

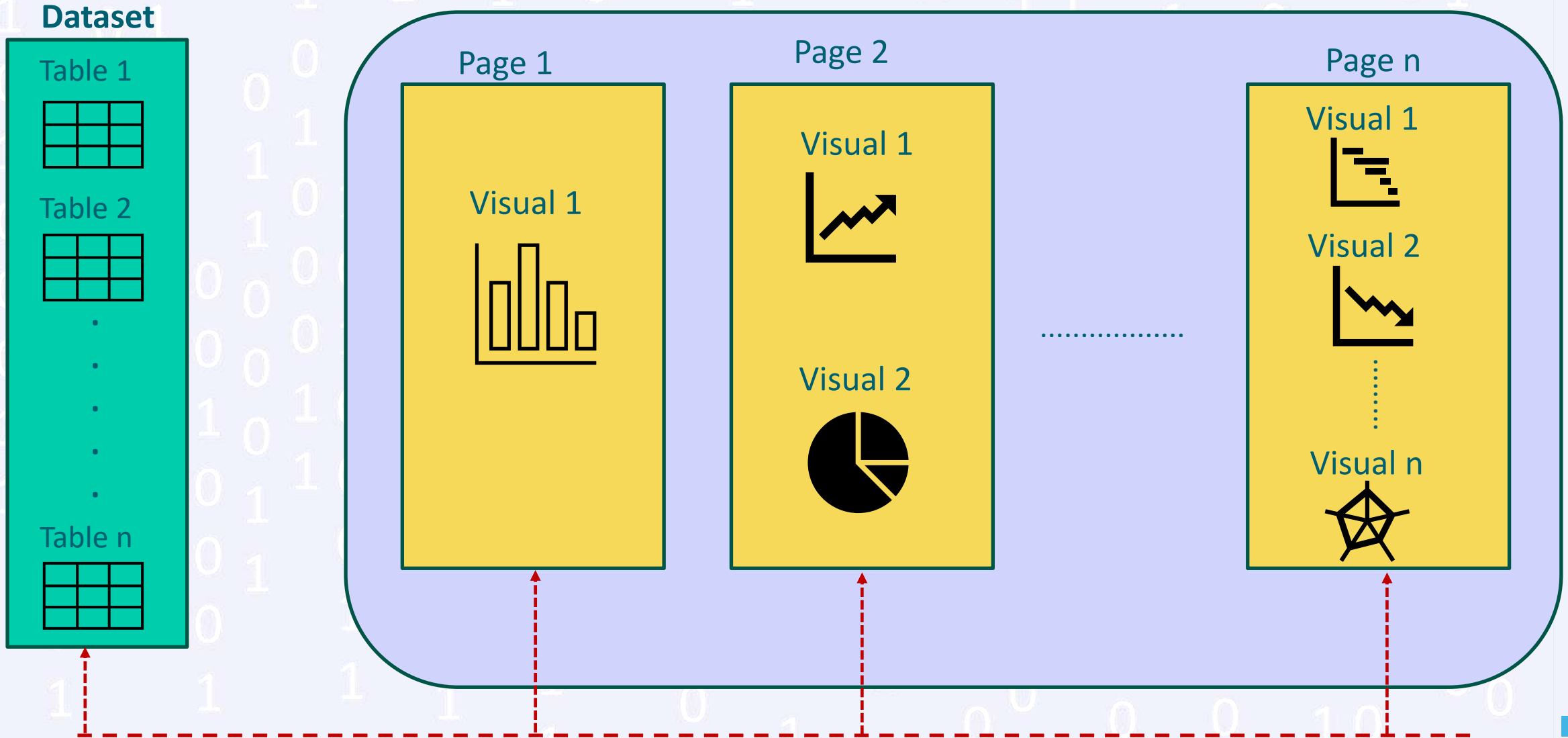
- Explore the different built-in visualizations in Power BI
- Choose the appropriate visuals for your report
- Understand the structure of a report & PBIX file

3.1.1 – Navigating the Power BI Desktop Report View

Report
view



3.1.2 – Report and PBIX File Structure in Power BI Desktop



3.1.3 – Power BI Built-in Visualizations

Traditional Charts

- Bar chart
- Column chart
- Line chart
- Area chart
- Pie chart

Geospatial visualizations

- Map
- Filled map
- Arc GIS

Esoteric Charts

- Combo chart
- Bubble chart
- Scatter plot
- Donut chart
- Gauge chart
- Funnel chart
- Waterfall chart

Text and Tables

- Single card
- Multi-row card
- KPIs
- Table
- Matrix

Other Visualizations

- Tree map
- Slicer
- Image

3.1.4 – What to Consider When Choosing a Type of Visualisation from the Gallery

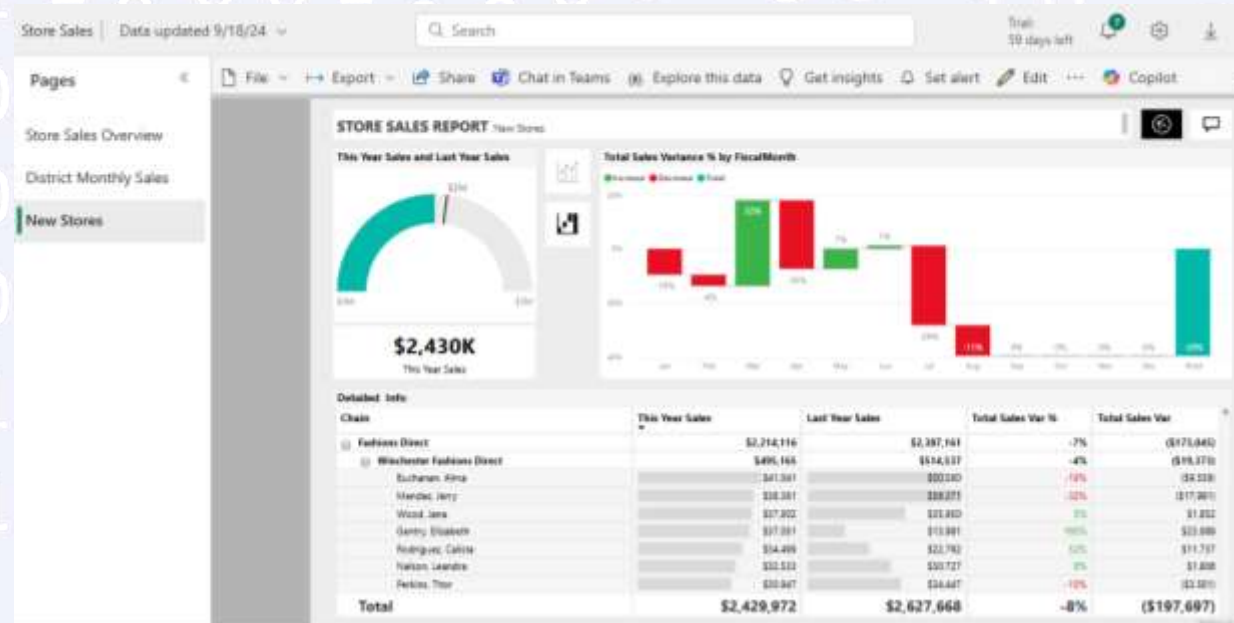
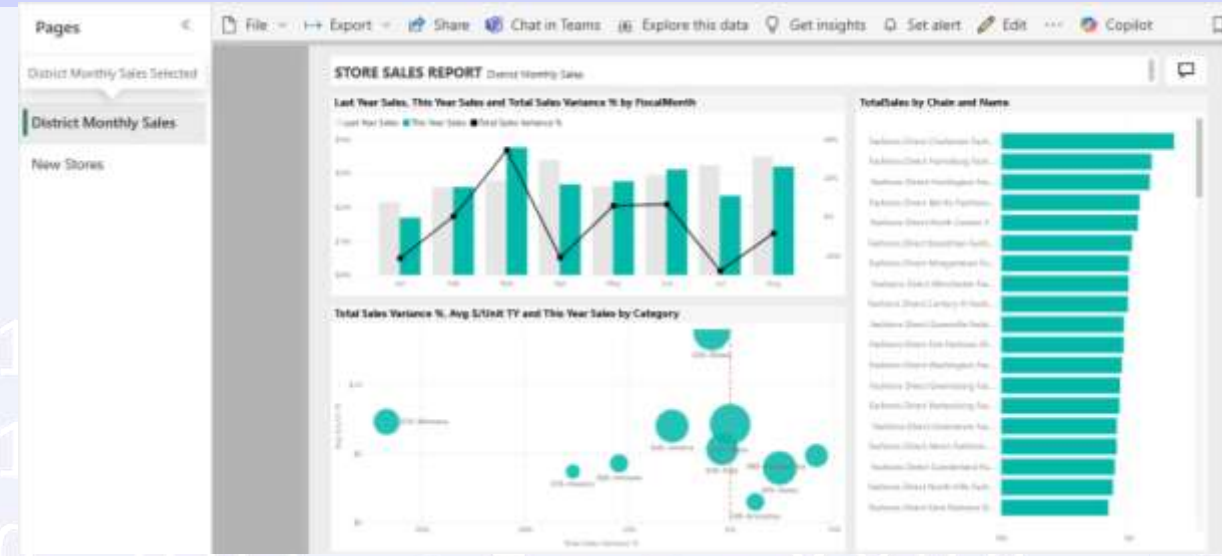
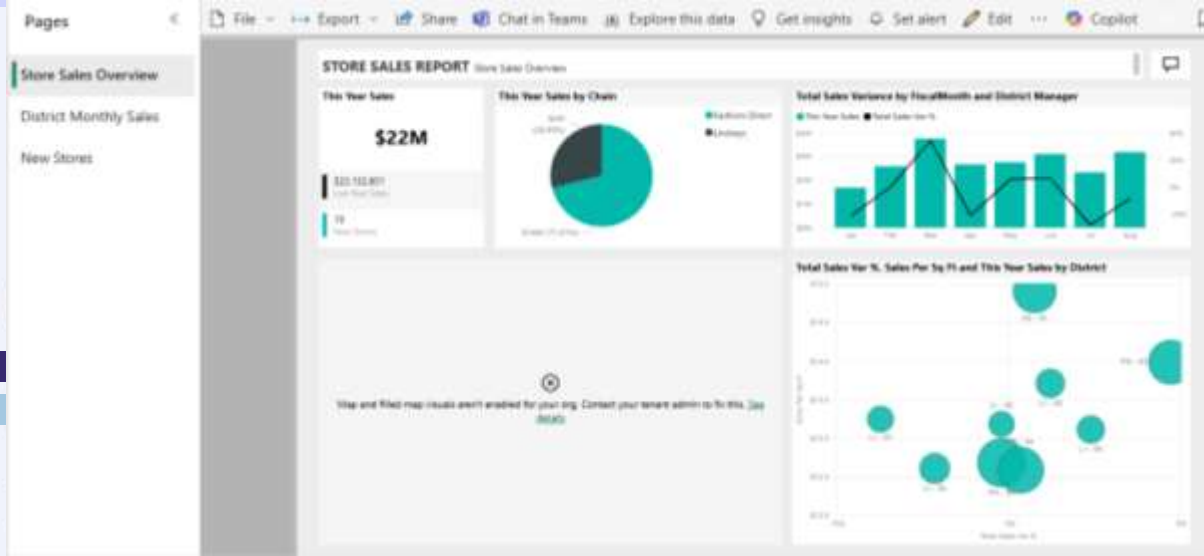
- Type of data you are dealing with (non-numerical, numerical, level of measurement)
- What you want to show
 - Composition of the data
 - Distribution of data series
 - Comparison among categories of a variable
 - Relationship between variables
- Your audience (Who)

Lesson 2 – Creating Reports in Power BI Desktop

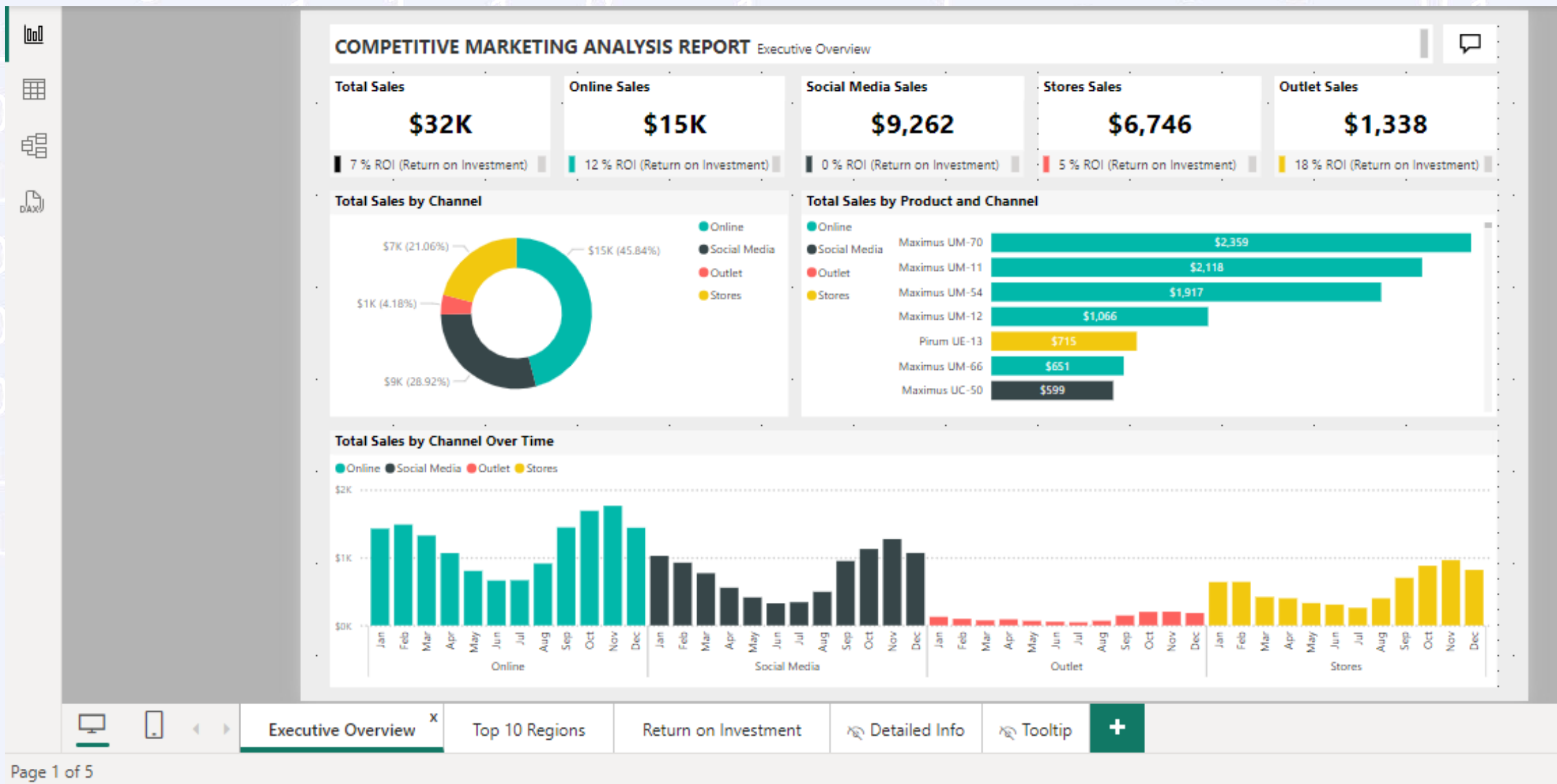
Objectives:

- Navigate a report in Power BI Desktop (being in the shoes of the consumers and end users)
- Plan the design and creation of a report
- Create immersive reports in Power BI Desktop

3.2.1 – Visualizing a Report in Power BI Desktop



3.2.2 – Navigating a Report in Power BI Desktop



3.2.3 – High Level Steps to Create a Report

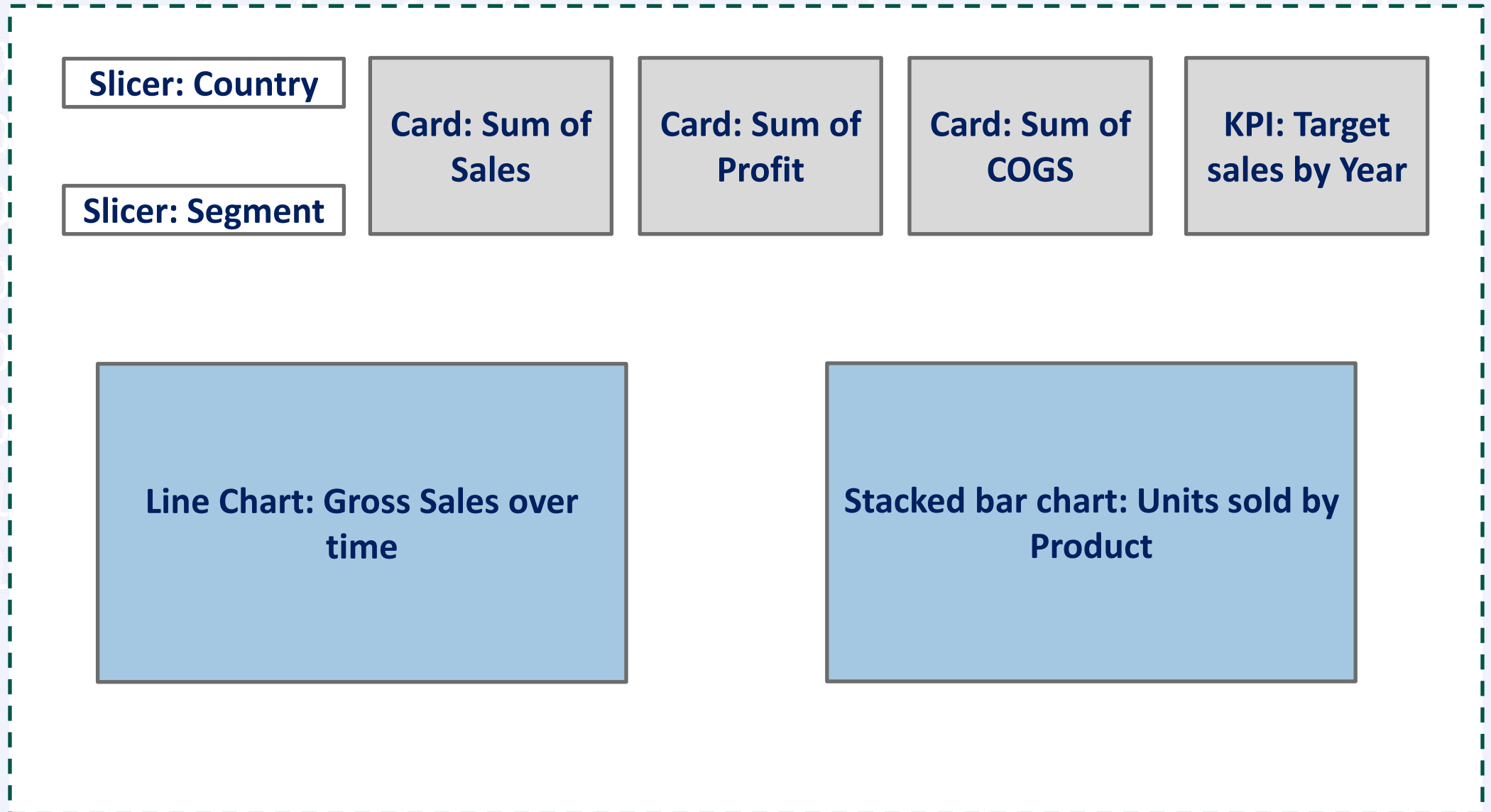
Planning Phase

- 1 Meeting with the different end users or consumers of the report
 - 2 Get an understanding of the KPIs that will be displayed in the report
 - 3 Identify the data source(s) that will feed the creation of the report.
 - 4 Collect the data needed for the creation of the dashboard.
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Production Phase

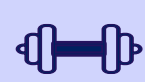
- 1 Clean, shape and transform your data
 - 2 Wireframe your report
 - 3 Explore the data tables and fields you have available
 - 4 Create a model if necessary or make sure your data tables are appropriately joined
 - 5 Polish your report with formatting and other tools in Power BI Desktop
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3.2.4 – Wireframing a Report



3.2.5 – Creating a New Report: Main Tasks

- Explore the visualizations gallery
- Add and arrange visualizations
- Create different types of visualizations (chart, text, map, and gauge)
- Add a slicer to filter visualizations
- Sort, copy and paste visualizations
- Add a filter to a page
- Add a filter to a report



Practice Exercise

- Visualizing a Power BI Desktop report: 321_Visualize Reports.pbix
- Navigating a report in Power BI Desktop: 322a_Navigate Report.pbix; 322b_Navigate Report.pbix

Lesson 3 – Formatting Reports

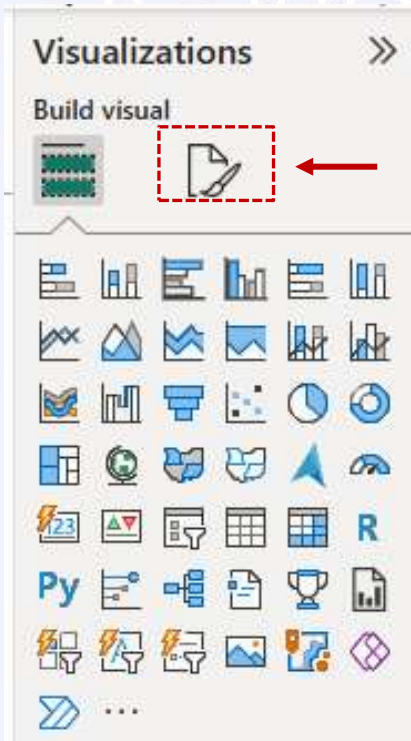
Objectives:

- Learn the formatting tools available in Power BI Desktop
- Format a report for better user experience
- Enhance reports for usability and storytelling

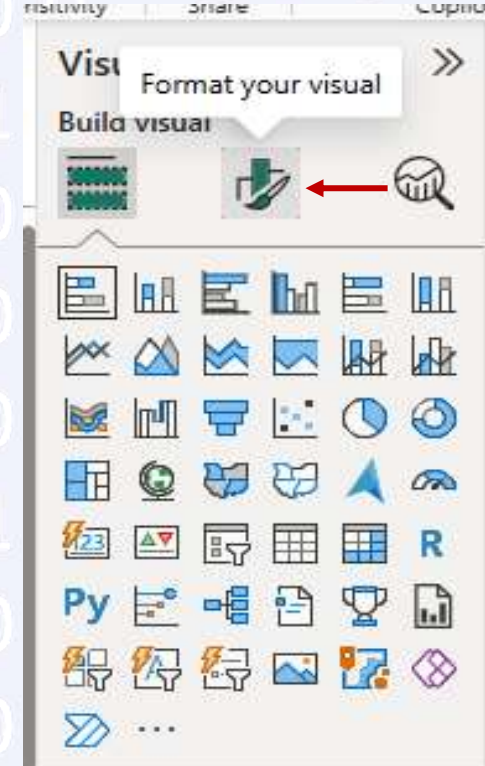
3.3.1 – Formatting Tools in Power BI Desktop

Formatting Tools

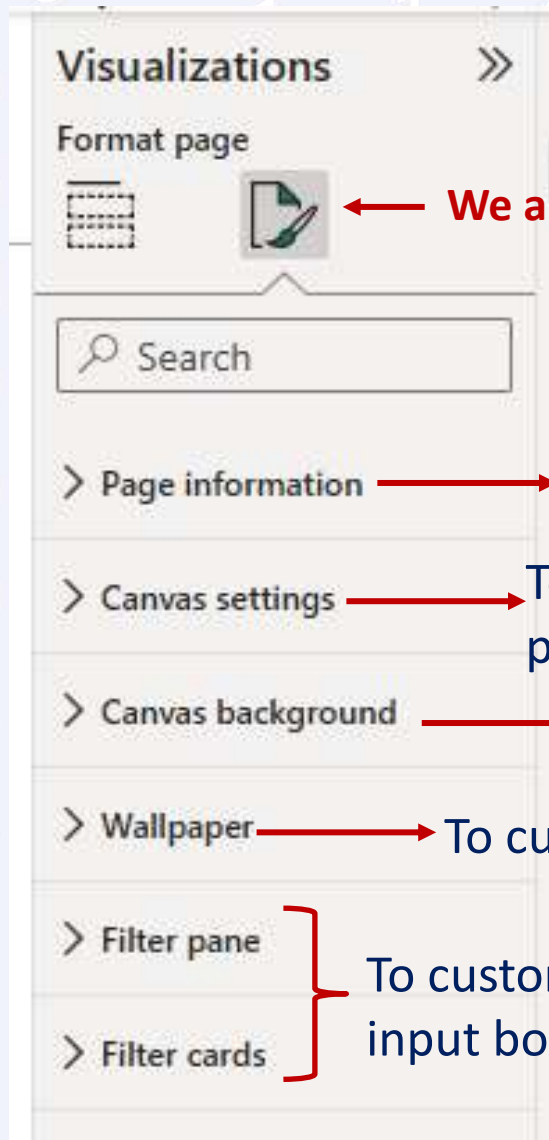
Report Page Formatting Tools



Visual Formatting Tools



3.3.2 – Formatting a Report Page



We are here

To name the report page. The page name enables it as a tooltip or allows it to be used for Q&A visuals.

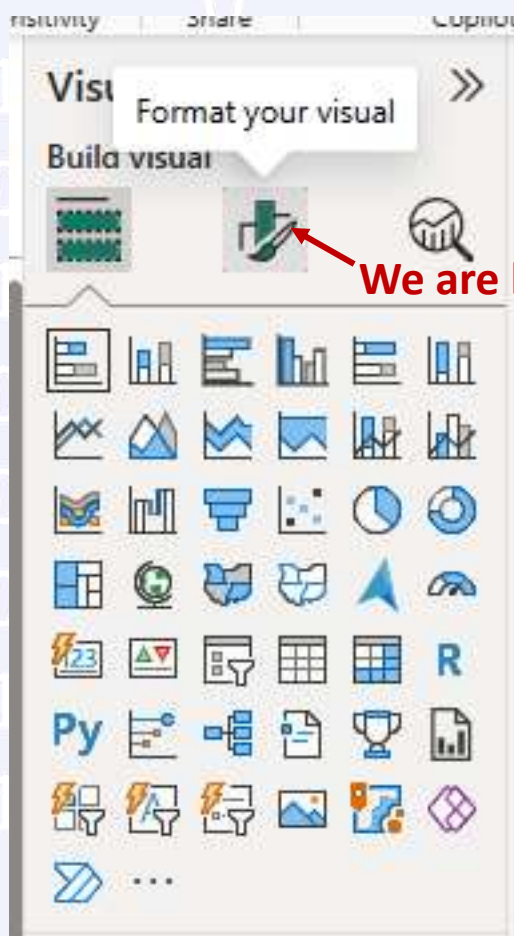
To set page size (e.g., 16:9, 4:3, Letter, or Tooltip) or define custom pixel dimensions.

To apply colors or images to the report area, with controls for transparency to improve readability.

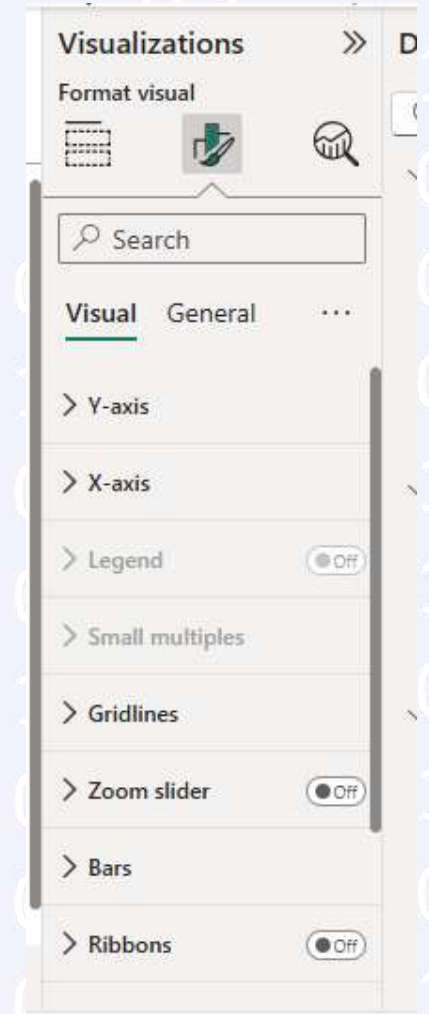
To customize the color or image behind the report canvas.

To customize formatting for the filters pane, including font, color, input box style, and background.

3.3.3 – Formatting Visuals



- Some formatting elements are specific to a particular visual
- All of the visuals have some common attributes (e.g., titles, location and size on the report page).
- The General tab provides some of the formatting elements you can make the same in different visualizations

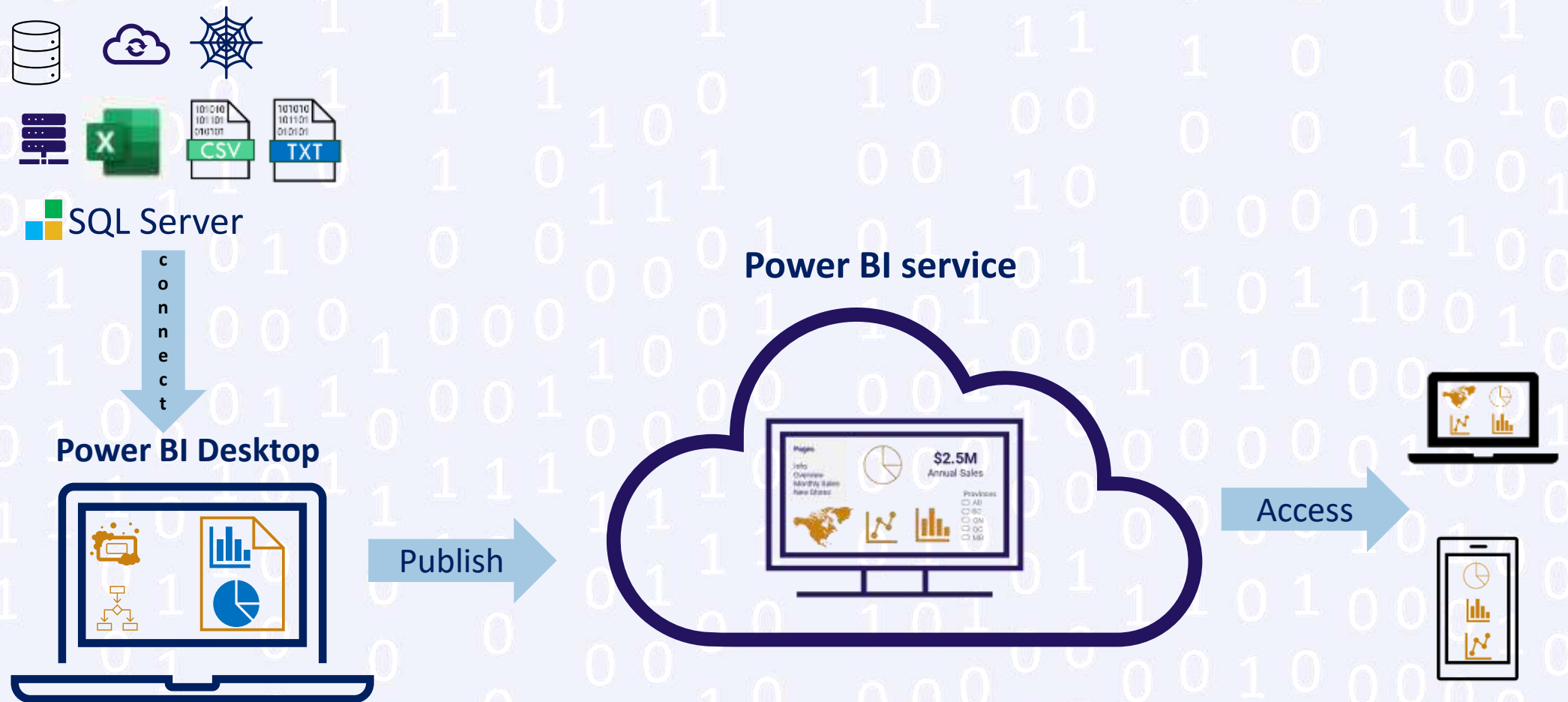


Lesson 4 – Publishing a Report to the Power BI Service

Objectives:

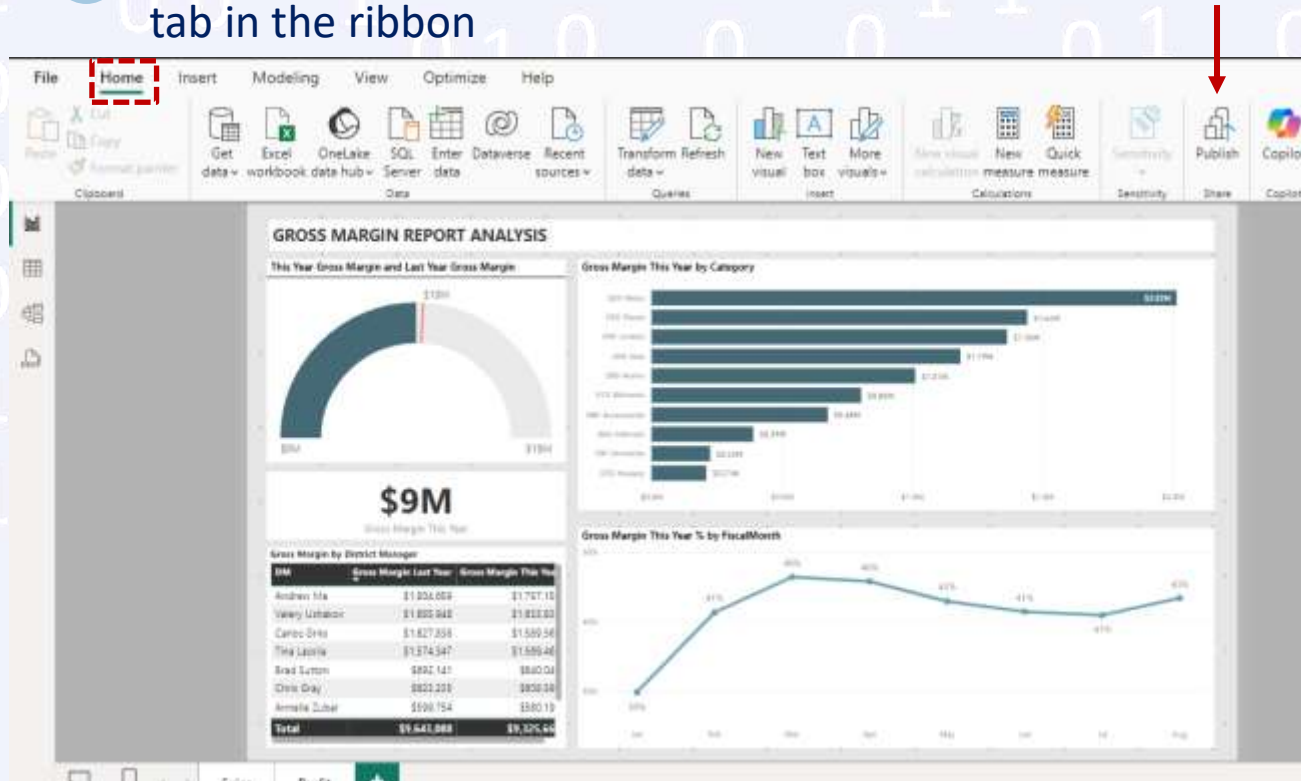
- Connect Power BI Desktop to Power BI Service
- Publish a Power BI Desktop report to the Power BI Service

3.4.1 – Connecting Power BI Desktop to Power BI Service

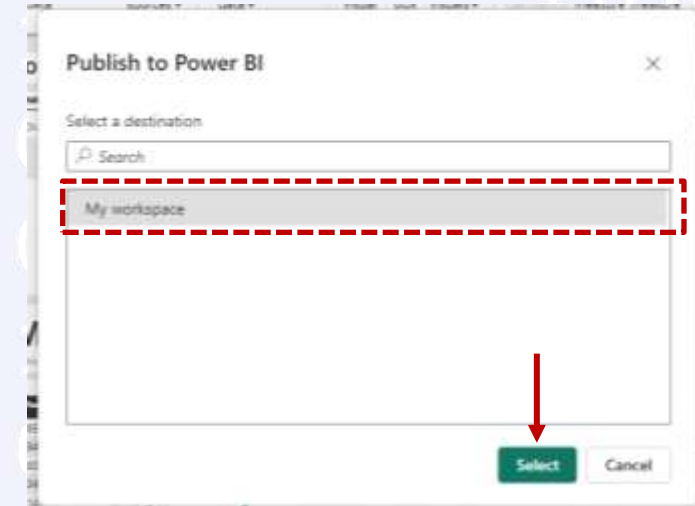


3.4.2 – Publishing a Power BI Desktop Report to the Power BI Service

- 1 Open your report in Power BI Desktop
- 2 Click or tap **Publish** on the **Home** tab in the ribbon



- 3 In the pop-up window, choose or specify the workspace where you want to publish your report. In this example, My workspace has been selected. Then, hit select at the bottom



- 4 Provide your credentials to connect to your Power BI service account
- 5 Open your Power BI account service to enjoy your report

Wrap-up